Business Questions

Introduction

This file is a series of 16 questions that will allow you to answer business problems.

Each question will follow the following pattern: ● Business question

- SQL query (you must insert a screenshot)
- Answer (you must insert a screenshot)

The first two queries and answers have been completed for you. Run the queries yourself to confirm you get the same answer and add your screenshots below.

Section 1: Basic Summary Statistics

Question 1: Unique Countries

How many unique countries are there?

SQL Query

```
SELECT count(distinct CountryName)
FROM PopStats;
```

Result

150

Question 2: Unique Cities

How many unique cities are there?

SQL Query

```
SELECT count(distinct City)
FROM FansPerCity
```

Result

46

Question 3: Unique Languages

How many unique languages are there?

SQL Query

```
SELECT COUNT(DISTINCT language) AS unique_languages FROM fansperlanguage;
```

Result(s)

41

Question 4: Global Page Reach

What is the daily average reach of the posts (i.e. DailyPostsReach) on the global page over the period?

SQL Query

```
WITH globaldaily AS (

    SELECT date, SUM(dailypostsreach) AS globaldailyreach

    FROM globalpage

    GROUP BY date

)

SELECT ROUND(AVG(globaldailyreach), 2) AS dailypostreach

FROM globaldaily;
```

Result(s)

83826721.36

Question 5: Global Page Likes

What is the daily average engagement rate (i.e. NewLikes) on the global page over the period? Round the result to 2 decimal places.

SQL Query

```
WITH globaldailylikes AS (

  SELECT date,

        SUM(newlikes) AS globaldailynewlikes
```

```
  FROM globalpage

  GROUP BY date

)

SELECT ROUND(AVG(globaldailynewlikes), 2) AS daily_average_new_likes

FROM globaldailylikes;
```

Result(s)

402415.04

Section 2: Location Analysis

In these queries, make sure you select the latest date in the main table. E.g:

SELECT max(date) from FansPerCountry

Then use the result of this in the WHERE clause of your main query.

Question 6: Top 10 countries by Number of Fans

What are the top 10 countries (considering the number of fans)? Show a table of results containing the following columns:

- CountryCode
- Country Name
- NumberOfFans

SQL Query

```
 SELECT     fanspercountry.countrycode,
popstats.countryname,
fanspercountry.numberoffans
FROM fanspercountry
JOIN popstats on fanspercountry.countrycode=popstats.countrycode
WHERE date = (SELECT MAX(date) FROM fanspercountry)  ORDER
BY
  numberoffans DESC LIMIT 10;
```

Result(s)

countrycode	countryname	numberoffans
CI	Ivory Coast	112160
CM	Cameroon	102211
SN	Senegal	83561
FR	France	73252
MG	Madagascar	72956
CD	Democratic Republic of the Congo	50705
BF	Burkina Faso	43500
ML	Mali	40578
DZ	Algeria	39093
GN	Guinea	36821

Question 7: Top 10 countries by Penetration Ratio

What are the top 10 countries by penetration ratio (i.e. the % of the country population that are fans) ? Show a table of results containing the following columns :

- CountryName
- PenetrationRatio
- NumberOfFans
- Population

SQL Query

```
SELECT
popstats.countryname,
   ROUND ((fanspercountry.numberoffans * 100.0) / popstats.population, 2) AS Penetration_Ratio,
fanspercountry.numberoffans,     popstats.population
FROM fanspercountry
JOIN popstats ON fanspercountry.countrycode = popstats.countrycode
WHERE date = (SELECT MAX (date) FROM fanspercountry)
ORDER BY Penetration_Ratio DESC
LIMIT 10;
```

Result(s)

countryname	penetrationratio	numberoffans	population
Reunion	2.41	20885	866506
French Polynesia	1.82	5148	283007
New Caledonia	1.79	5032	280460
Mauritius	1.77	24210	1364283
Martinique	1.44	5427	376480
Guadeloupe	1.36	5379	395700
Gabon	1.13	23954	2119036
Mayotte	0.73	1983	270372
Comoros	0.60	4925	821164
French Guiana	0.57	1687	296711

Question 8: Bottom 10 Cities by Number of Fans

What are the bottom 10 cities (considering the number of fans) among countries with a population over 20 million? This could be considered our growth potential.
Show a table of results containing the following columns:

- CountryName
- City
- NumberOfFans
- Population

SQL Query

```
SELECT
popstats.countryname ,
fanspercity.city ,
   SUM(fanspercity.numberoffans ) AS NumberOfFans,     popstats.population
FROM fanspercity
JOIN popstats ON fanspercity.countrycode = popstats.countrycode
WHERE
    (popstats.population > 20000000 and  date = (select max(date) from fanspercity))
GROUB BY        popstats.countryname ,fanspercity.city,popstats.population  ORDER
BY        sum(fanspercity.numberoffans )ASC LIMIT 10;
```

Result(s)

countryname	city	numberoffans	population
Algeria	Bejaia	2301	41657488
Cameroon	Ngaoundere	2318	25640965
Madagascar	Fianarantsoa	2366	25683610
Algeria	Tizi Ouzou	2524	41657488
Canada	Montreal	2887	35881659
Algeria	Oran	2920	41657488
Ivory Coast	Bouake	3376	24290000
Ivory Coast	Cocody	3647	24290000
Morocco	Casablanca	3951	34314130
Angola	Luanda	4614	30355880

Section 3: Fan Analysis

Question 9: Analysis by age group (split of fans)

What is the split of page fans across age groups (in %)?
Show a table of results containing the following columns:

- AgeGroup
- PercentageOfFans

SQL Query

```
SELECT agegroup,
ROUND(
    SUM(numberoffans)*100.00/
    (SELECT SUM(numberoffans) from fanspergenderage), 2 ) AS PercentageOfFans
FROM fanspergenderage  GROUB
BY
   agegroup
ORDER BY
   PercentageOfFans desc;
```

Result(s)

Agegroup	percentageoffans
	35.73

25-34	
18-24	21.30
35-44	19.40
45-54	9.47
65+	6.94
55-64	5.02
13-17	2.13

Question 10: Analysis by gender (split of fans)

What is the split of page fans by gender (in %)?

Show a table of results containing the following columns:

- Gender
- PercentageOfFans

SQL Query

```
SELECT gender,
ROUND(
    SUM(numberoffans)*100.0/
    (SELECT SUM(numberoffans) FROM fanspergenderage),2)
AS PercentageOfFans
FROM fanspergenderage  GROUB
BY
    gender
ORDER BY
    PercentageOfFans;
```

Result(s)

Gender	percentageoffans
U	0.09
M	43.44
F	56.46

Section 4: Language Analysis

Question 11: English Speaking Fans

What is the number of the fans that have declared English as their primary language ?

SQL Query

```
SELECT SUM(numberoffans)  AS English_Fans FROM fansperlanguage  WHERE language = 'en';
```

Result(s)

English_fans

1347752

Question 12: English Speaking Fans Percentage

What is the percentage of the fans that have declared English as their primary language?

SQL Query

```
SELECT SUM(numberoffans) as English_Fans,
ROUND(
    SUM(numberoffans)*100.0/
    (SELECT SUM(numberoffans) FROM fansperlanguage),2)
AS   English_Fans_Percentage
FROM fansperlanguage
WHERE
     LANGUAGE= 'en' ;
```

Result(s)

english_fans	english_	fans_percentage
1347752		5.08

Question 13: Buying Power of English Speakers

Based on the number of fans who have declared English as their primary language and living in the US, what is the potential buying
power that can be accessed? (Please use the average income data per country for this question. It is estimated that on average,
0.01% of the annual income is dedicated to online magazine subscriptions in the US).

SQL Query

```
SELECT
   ROUND(
```

```
        SUM(f.NumberOfFans * p.AverageIncome * 0.0001), 2)
        AS PotentialBuyingPower
FROM FansPerLanguage f
JOIN
      PopStats p ON f.CountryCode = p.CountryCode
WHERE
      f.Language = 'en'    AND p.CountryName = 'United states';
```

Result(s)

potentialbuyingpower
5464504.63

Section 5: Fan Engagement

Question 14: Engagement per day of the week

What is the split of the EngagedFans across the days of the week (monday, tuesday,...)?

Give the result as a table with the following columns:

- DayOfWeek
- PercentageSplit

Based on the results, what is the best day of the week to publish posts?

SQL Query

```
 SELECT
   TO_CHAR(createdtime::timestamp, 'Day') AS dayofweek,
   ROUND(
      SUM(engagedfans) * 100.0 /
      (SELECT SUM(engagedfans) FROM postinsights), 2 ) AS percentagesplit
FROM postinsights GROUP BY          dayofweek
ORDER BY
      percentagesplit DESC;
```

Result(s)

dayofweek	percentagesplit
Saturday	19.73
Monday	19.23
Tuesday	18.67
Wednesday	15.38
Sunday	12.08
Friday	8.59

Thursday	6.32

Question 15: Engagement per time of day

What is the split of the EngagedFans by time of the day? Split the day into the following time ranges (call the column TimeRange):

- 05:00 - 08:59
- 09:00 - 11:59 ● 12:00 - 14:59 ● 15:00 - 18:59 ● 19:00 - 21:59 ● 22:00 or later

Give the result as a table with the following columns:

- TimeRange
- Precentage

Based on the results, what is the best time of the day to publish posts?

SQL Query

```
WITH time_bucket AS (
 SELECT
CASE
WHEN EXTRACT(HOUR FROM createdtime::timestamp) BETWEEN 5 AND 8
        THEN '05:00 - 08:59'
WHEN EXTRACT(HOUR FROM createdtime::timestamp) BETWEEN 9 AND 11
        THEN '09:00 - 11:59'
WHEN EXTRACT(HOUR FROM createdtime::timestamp) BETWEEN 12 AND 14
        THEN '12:00 - 14:59'
WHEN EXTRACT(HOUR FROM createdtime::timestamp) BETWEEN 15 AND 18
        THEN '15:00 - 18:59'
WHEN EXTRACT(HOUR FROM createdtime::timestamp) BETWEEN 19 AND 21
        THEN '19:00 - 21:59'
ELSE '22:00 or later'
      END AS TimeRange,
                Engagedfans   FROM postinsights
)
SELECT  timerange,
   ROUND(
    SUM(engagedfans) * 100.0 /(SELECT SUM(engagedfans) FROM postinsights),2)
AS Percentage
FROM time_bucket
GROUP BY TimeRange
ORDER BY Percentage DESC;
```

Result(s)

timerange	percentage
05:00 - 08:59	33.96

15:00 - 18:59	17.40
09:00 - 11:59	15.34
19:00 - 21:59	13.12
12:00 - 14:59	12.70
22:00 or later	7.48

Section 6: Optional Challenging Queries

Question 16: Month to month change in engagement

Compute the change in PostClicks, EngagedFans and Reach from one month to the next. Give the result as a table with the following columns:

- FromMonth
- ToMonth
- DeltaPostClicks
- DeltaEngagedFans
- DeltaReach

Where the deltas are the percentage differences between a month and its previous month.

SQL Query

Paste query here

Result(s)

Copy and paste result(s) of query or paste screenshot of your result(s) here